Financial Conditions, r^*, and the Transmission of Monetary Policy Surprises

Draft for discussion

1 November 2026

Abstract

High-frequency surprises identify monetary policy shocks cleanly, but their estimated effects on unemployment and inflation after 1994 are often weak or puzzling. We hold the surprise fixed and change only how it is propagated. When the surprise reaches the economy through a nominal short rate alone, the familiar puzzles appear. When it propagates through financial conditions as a whole, measured with rates relative to the Laubach–Williams neutral rate r^*, unemployment rises by about 0.3 percentage point within a year per point on the two-year yield and the core price level falls about 0.5 percent after two years, close to the best recent estimates. The financial conditions index is essential. Removing the slow-moving component of rates matters for precision; the significant price response depends on how that component is measured and on data since 2021.

1 Introduction

The high-frequency identification of monetary policy shocks is one of the major empirical advances in macroeconomics. Changes in interest-rate futures and Treasury yields in narrow windows around FOMC announcements isolate news about policy from other news (Kuttner, 2001; Gürkaynak et al., 2005), and their effects on asset prices are estimated precisely (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021). Using these surprises to learn about the effects of policy on output, unemployment and inflation has been harder. In samples that start in the 1990s, the estimated responses are often small, imprecise or of the wrong sign (Ramey, 2016). The literature has made major progress on this problem. One line of work shows that announcements also reveal the Federal Reserve's information about the economy, so that a surprise tightening partly signals good news, and separates the two components (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). Another shows that surprises are partly predictable from public data that predate the announcement, and that orthogonalizing them, and adding Chair speeches, restores conventional and larger responses (Bauer and Swanson, 2023a,b). Closest to us, Gertler and Karadi (2015) show that policy surprises work through credit costs, and Caldara and Herbst (2019) that omitting credit spreads, to which policy also reacts, attenuates the response of activity. The link between futures-based surprises and monthly VARs goes back to Faust et al. (2004).

This note focuses on the other half of the exercise: how the surprise is propagated to the economy. We hold the high-frequency shock fixed and ask how much of the post-1994 puzzle reflects two features of the transmission mechanism that standard specifications leave out. First, policy affects spending through the whole configuration of financial conditions—the cost of long-term credit, the value of wealth and the exchange rate (Boivin et al., 2010)—not through a single

short rate; summarizing them in a financial conditions index is common in policy analysis (Hatzius et al., 2010; Ajello et al., 2023). Second, what matters for spending and inflation is the level of real rates relative to the neutral rate r^*, which fell substantially from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017; Hamilton et al., 2016; Kiley, 2020) and shaped the trend in yields (Bauer and Rudebusch, 2020). Over the post-1994 period, rates trended down with r^* while activity trended up; a reduced-form link between the level of rates and activity then partly reflects these opposing trends rather than the cyclical stance of policy.

Our design is direct. The shock is the Nakamura and Steinsson (2018) policy-news surprise, measured in 30-minute windows (253 FOMC announcements, 1994–2026). Its impact on the two-year yield and, when included, on a market-based financial conditions index (FCI) is read off the window, where macroeconomic variables do not move. A daily model carries the financial variables forward from the announcement day, and monthly unemployment and core inflation respond to monthly averages of the daily variables, as in a standard monthly VAR. The model respects the timing of each announcement within its month, which addresses the temporal-aggregation problem stressed by Kilian (2024) and Chudik and Georgiadis (2022); related mixed-frequency approaches include Francis et al. (2011), Ghysels (2016) and Gazzani and Vicondoa (2020). The FCI and the sequence of steps—nominal, real, and relative to r^*—are taken from a companion note on financial conditions (FCI note, 2026) and were fixed before this analysis.

The evidence builds in four steps. With the nominal two-year yield as the only channel, the post-1994 puzzles appear: unemployment falls after a tightening and prices do not move. Adding the FCI turns the unemployment response conventional, but it is imprecise at short horizons and sensitive to the lag length. Measuring rates in real terms changes little. Measuring rates and the FCI relative to r^* gives an unemployment response that is significant at horizons of one to three years for every lag length and nearly invariant to it, and a core price level that falls significantly with 6 and 12 lags. Removing r^* without the FCI does not fix the unemployment response. The magnitudes are close to the benchmark of Bauer and Swanson (2023a): unemployment rises about 0.3 percentage point in the first year against their 0.2, and stays higher for longer; the price level falls by about 0.5 percent after two years, as in their estimates.

The robustness exercises sharpen what drives this. The FCI is essential in every variant we consider. The slow-moving component of rates must be removed, but for unemployment simple statistical trends do about as well as r^*; the significant price response, by contrast, requires a model-based estimate of r^* or a smooth trend, and the inclusion of data since 2021. Excluding 2021, or ending the sample in 2019, the unemployment response remains conventional and significant at short horizons but is smaller. Kiley's estimate of r^* and a smooth time trend give similar results; a backward-looking average of the two-year yield and a market-based proxy, the 9–10-year forward rate, work less well, especially for prices. With industrial production in the system, the FCI alone already delivers a significant unemployment response.

These results are complementary to the information-effect and predictability literatures. Our propagation is estimated from the reduced-form link between financial conditions and the economy, which we argue makes it less exposed to the correlation between surprises and prior news emphasized by Bauer and Swanson (2023a); and the surprise reaches macro variables only through financial conditions, which screens out information channels that bypass markets (information that moves equity prices in the window is still included). Combining orthogonalized or decomposed surprises with propagation through financial conditions relative to r^* is a natural next step. To our knowledge, propagating high-frequency surprises through a broad financial conditions index measured relative to r^* has not been examined.

2 Data and method

Surprises and window responses. The shock is the Nakamura–Steinsson policy-news factor from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025): the first principal component of 30-minute changes in five short-term interest-rate futures around 253 FOMC announcements, February 1994 to September 2026. We scale it so that the two-year Treasury yield rises 1 percentage point in the window. The window response of the FCI is computed from the 30-minute changes in the 10-year yield, the S&P 500 and the dollar index in the same database, combined with the FCI's weights; it is 0.99 per percentage point, close to one by construction, since the weights are scaled so that a unit of the index corresponds to a policy move that raises the two-year yield 1 percentage point. Unemployment and prices are zero in the window.

Financial conditions. The FCI is the market index of FCI note (2026): a weighted sum of the 10-year Treasury yield, equity valuation (cyclically adjusted price–earnings ratio, CAPE) and the real broad dollar, with weights proportional to their responses to the same policy news. It comes in three versions, built in the same steps as in that note. *Nominal*: the variables as measured. *Real*: the 10-year yield net of 10-year expected inflation. r^**-gap* (baseline): the real 10-year yield net of the one-sided Laubach–Williams r^* (New York Fed, current estimates), and equities as the gap between the earnings yield and r^*. The two-year yield follows the same rule: nominal; net of two-year expected inflation (Cleveland Fed; Haubrich et al., 2012); and net of expected inflation and r^*. Daily versions follow the daily 10-year yield, CAPE and dollar and are shifted, smoothly within each month, so that their monthly averages equal the monthly indexes exactly; the adjustments for expected inflation and r^* move slowly, so the window responses are the same in every version.

Model. The model has a daily and a monthly block. In the daily block, the two-year yield and (when included) the FCI follow a heterogeneous autoregression in their 1-, 5-, 22- and 66-day averages and respond to the current and previous month's unemployment gap and core inflation (estimated 1990–2026). In the monthly block, the unemployment gap (unemployment rate less the CBO natural rate) and core PCE inflation are rows of a monthly VAR with L lags of each and L lags of the monthly averages of the daily financial variables, estimated by OLS over February 1994–September 2026 with March–December 2020 excluded as dependent observations; L is 3, 6 or 12. Within each month macro conditions are set first, and markets respond to them; in the announcement month, macro variables have not yet moved. The response to a surprise is computed by placing the window impact on the announcement day (drawn from the empirical distribution of FOMC dates within the month), simulating the daily block forward, averaging within months, and iterating the monthly block. This is a restricted mixed-frequency VAR (Ghysels, 2016) with equal weights within the month, the restriction implied by a standard monthly VAR. A test of equal within-month weights does not reject it for unemployment; it is rejected for core inflation, although the gain in fit is very small.

Identifying assumption. The surprise affects unemployment and prices only through the variables in the daily block, and the economy responds to a given change in those variables in the same way whatever caused it. The first part is weaker the more complete the set of financial variables; the second is the assumption behind any use of financial conditions indexes to summarize the stance of policy. It is strong: equity prices, for example, partly anticipate activity. A direct mixed-frequency regression of unemployment on the surprises that does not impose the assumption (Chudik and Georgiadis, 2022; Chudik and Kilian, 2026) gives wrong-signed point estimates, driven largely by the 2007–09 announcements, and is about ten times less precise than our estimates; it cannot discriminate between them and zero.

Inference. We resample announcements (which re-estimates the window impact and the distribution of announcement dates) and residuals of the monthly equations, 200 draws; the daily

block, the FCI weights and r^* are held fixed, so the intervals understate uncertainty. We report 90 percent intervals, the convention in recent high-frequency studies (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016), and also 68 percent intervals, which Jarociński and Karadi (2020) report alongside 90 percent ones; Gertler and Karadi (2015) use 95 percent.

3 Results

Tables 1 and 2 report the responses of the unemployment gap and the core PCE price level at 12, 24 and 36 months, for 3, 6 and 12 monthly lags, at each step. Figure 1 shows the four key cases.

Table 1: Unemployment gap (percentage points), per 1 percentage point policy-induced rise in the two-year yield

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	$-0.13^{\dagger}$	-0.27^{**}	-0.34^{**}	-0.01	$-0.19^{\dagger}$	-0.36^{**}	0.08	-0.08	$-0.25^{\dagger}$
Nominal	yes	0.18^{**}	0.09	0.07	0.38^{**}	0.29^{**}	$0.19^{\dagger}$	0.42^{**}	0.37^{**}	0.24^{**}
Real	no	-0.02	$-0.14^{\dagger}$	-0.25^{**}	0.07	-0.10	-0.28^{**}	0.12	-0.03	$-0.21^{\dagger}$
Real	yes	0.25^{**}	$0.18^{\dagger}$	0.12	0.40^{**}	0.33^{**}	0.24^{**}	0.38^{**}	0.36^{**}	0.26^{**}
r^*-gap	no	0.08	0.03	-0.08	$0.16^{\dagger}$	0.04	$-0.15^{\dagger}$	$0.20^{\dagger}$	0.05	-0.16
r^*-gap	yes	0.30^{**}	0.32^{**}	0.34^{**}	0.38^{**}	0.39^{**}	0.41^{**}	0.34^{**}	0.38^{**}	0.37^{**}

Columns: horizon and number of monthly lags. **: 90 percent interval excludes zero; †: 68 percent interval excludes zero (90 percent does not). Rates and FCI at each step as described in Section 2. Bauer and Swanson (2023a), benchmark: +0.19 at 12 months (90 percent band excludes zero), +0.17 and +0.15 at 24 and 36 months (band includes zero), read from their Figure 8.

Table 2: Core PCE price level (percent), per 1 percentage point policy-induced rise in the two-year yield

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	0.01	0.06	0.01	-0.08	0.03	0.05	-0.17	-0.07	0.03
Nominal	yes	-0.09	-0.07	-0.10	$-0.28^{\dagger}$	$-0.24^{\dagger}$	-0.23	$-0.42^{\dagger}$	$-0.41^{\dagger}$	$-0.36^{\dagger}$
Real	no	$-0.10^{\dagger}$	-0.02	-0.07	$-0.19^{\dagger}$	-0.08	-0.06	$-0.26^{\dagger}$	-0.16	-0.08
Real	yes	$-0.17^{\dagger}$	$-0.14^{\dagger}$	-0.18^{**}	$-0.33^{\dagger}$	$-0.32^{\dagger}$	$-0.31^{\dagger}$	$-0.44^{\dagger}$	-0.47^{**}	$-0.41^{\dagger}$
r^*-gap	no	$-0.15^{\dagger}$	$-0.13^{\dagger}$	-0.19^{**}	$-0.22^{\dagger}$	$-0.26^{\dagger}$	$-0.31^{\dagger}$	$-0.28^{\dagger}$	-0.39	$-0.44^{\dagger}$
r^*-gap	yes	$-0.18^{\dagger}$	-0.25^{**}	-0.33^{**}	$-0.28^{\dagger}$	-0.48^{**}	-0.61^{**}	$-0.37^{\dagger}$	-0.68^{**}	-0.84^{**}

See Table 1. Bauer and Swanson (2023a), benchmark for the CPI (not core PCE): −0.36, −0.48 and −0.6 at 12, 24 and 36 months, 90 percent bands exclude zero.

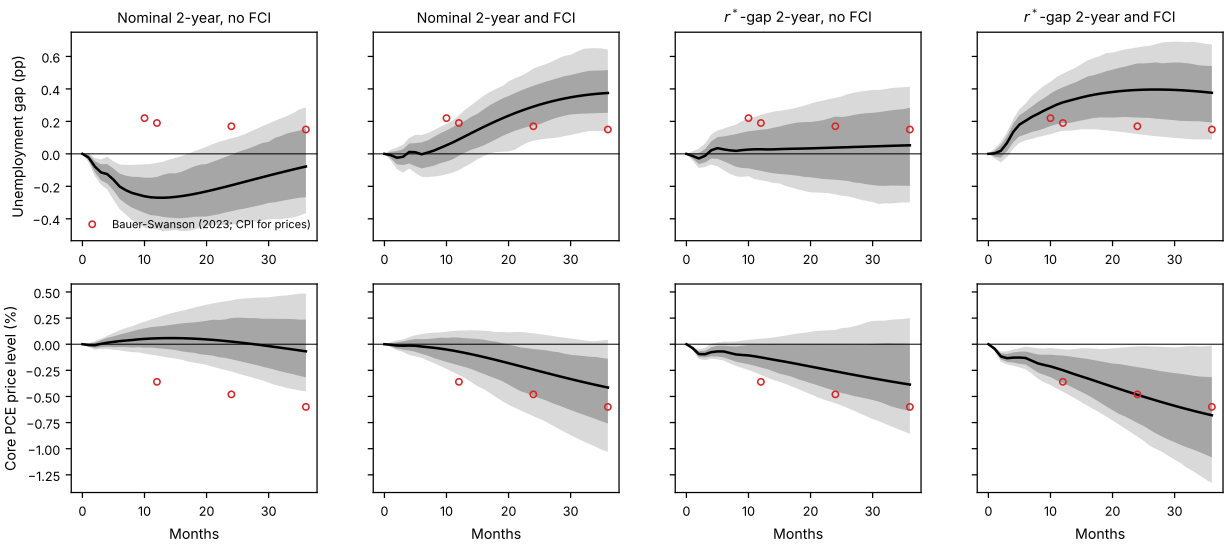


Figure 1: Responses to a policy move that raises the two-year yield 1 percentage point (6 lags). Dark band: 68 percent; light band: 90 percent. Circles: Bauer and Swanson (2023a), read from their Figure 8 (unemployment rate; CPI rather than core PCE).

Step 1: the standard set-up reproduces the puzzles. With the nominal two-year yield as the only channel—the policy indicator of Bauer and Swanson (2023a)—unemployment falls after a tightening, significantly with 6 and 12 lags, and the price level does not move. This is the post-1994 pattern documented by Ramey (2016) and the one Bauer and Swanson (2023a) obtain with unadjusted surprises.

Step 2: financial conditions. Adding the FCI turns the unemployment response positive at every horizon and lag length. It is significant at 36 months, but at 12 months it is weak with 6 and 12 lags. The price level falls by 0.2–0.4 percent at 24–36 months; the 90 percent intervals include zero (the 68 percent intervals exclude it at 36 months for every lag length).

Step 3: real rates. Measuring the two-year yield and the FCI net of expected inflation changes little, as in the companion note, where the nominal and real indexes are close after 1990.

Step 4: rates relative to r^*. With the FCI, measuring the two-year yield and the FCI relative to r^* gives an unemployment response of 0.30–0.34 percentage point at 12 months and 0.34–0.38 at 36 months, significant at the 90 percent level at every horizon and lag length and nearly identical across lag lengths. The price level falls by 0.48–0.61 percent at 24 months with 6 and 12 lags, significant at the 90 percent level (with 3 lags the 68 percent interval excludes zero). Without the FCI, the r^*-gap rate leaves unemployment insignificant; the price level falls somewhat (68 percent intervals exclude zero at most horizons).

Comparison with Bauer and Swanson (2023a). Their benchmark proxy VAR (1973–2020, with surprises from FOMC announcements and Chair speeches, orthogonalized to pre-announcement news) gives a peak unemployment response of about 0.2 percentage point after 10 months and a CPI decline of about 0.5 percent after two years, per percentage point on the two-year yield. Our step-4 estimates are similar for prices (core PCE rather than the CPI) and give a larger and more persistent unemployment response, with intervals that exclude zero through 36 months where theirs include zero after about a year and a half. Their estimates use 30 years of instrument data, Chair speeches and orthogonalization; ours use FOMC windows alone and no orthogonalization. Earlier versions of our model with Greenbook-purged surprises gave nearly identical results.

4 Robustness

Industrial production. Table 3 adds industrial production to the monthly block. Without the FCI, industrial production rises after a tightening at every step; with the FCI it falls; at the r^* step it falls about 0.9 percent at 12 months, significantly with 6 and 12 lags, against about 1.5 percent in Bauer and Swanson (2023a), and recovers faster. The unemployment results differ from the core system at steps 1 and 2: with industrial production included, the step-1 puzzle is much weaker, and the nominal FCI already gives a significant unemployment response at 12 months (0.25–0.41). The price results are as in the core system. In exploratory work, adding a commodity price index as well produced rising commodity prices and industrial production after a tightening, a reminder that the propagation assumption is least plausible for the most forward-looking, demand-sensitive variables.

Table 3: Monthly block with industrial production: unemployment gap (top) and industrial production (bottom)

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	−0.01	−0.06	−0.20†	0.21†	0.16	−0.08	0.34†	0.34	0.16
Nominal	yes	0.41**	0.36**	0.25**	0.63**	0.62**	0.52**	0.57**	0.61**	0.57**
Real	no	0.06	0.02	−0.14†	0.22†	0.16	−0.05	0.31†	0.26	0.11
Real	yes	0.41**	0.38**	0.27**	0.56**	0.56**	0.49**	0.47**	0.51**	0.51**
r^*-gap	no	0.10†	0.09	−0.03	0.20†	0.13	−0.05	0.25†	0.13	0.01
r^*-gap	yes	0.28**	0.34**	0.33**	0.35**	0.39**	0.41**	0.30**	0.37**	0.38†

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Nominal	no	1.15**	1.15**	1.64**	1.05†	1.10†	1.63**	1.01†	1.04†	1.23†
Nominal	yes	−0.17	−0.36	−0.08	−0.10	−0.14	−0.09	0.50	0.35	0.27
Real	no	0.70†	0.68†	1.13**	0.71†	0.83†	1.29**	0.76	0.92†	1.04†
Real	yes	−0.36	−0.59†	−0.36	−0.12	−0.20	−0.16	0.48	0.30	0.27
r^*-gap	no	0.22	0.15	0.43	0.26	0.56	0.87	0.33	0.82	0.77
r^*-gap	yes	−0.47†	−0.88**	−0.91**	−0.14	−0.32	−0.51	0.36	0.10	−0.12

See Table 1. Bauer and Swanson (2023a), industrial production: about −1.5, −1.0 and −0.85 at 12, 24 and 36 months, read from their Figure 8. Price-level responses (not shown) are as in Table 2.

Which trend? Table 4 holds the FCI at the three indexes of the companion note (nominal, real and r^*-gap) and varies only how the slow-moving component of the two-year yield is removed. Four results stand out. First, the FCI is essential in every variant: without it, unemployment is insignificant or significantly negative. Second, with the baseline Laubach–Williams adjustment of the two-year yield, all three FCIs give significant responses of both unemployment and prices; the r^*-gap FCI gives the most lag-robust ones. Third, Kiley (2020)'s estimate of r^* gives very similar results with the r^*-gap FCI: unemployment rises 0.27–0.37 percentage point at every horizon, significant at the 90 percent level, and the price level falls by 0.3–0.5 percent at 24–36 months, significant at the 68 percent level (and at 12 months at the 90 percent level with 6 and 12 lags). A quadratic time trend in every monthly equation, with the r^*-gap FCI, also gives significant responses of both unemployment (0.3–0.4 at 12 months, 0.5–0.6 at 24–36) and prices (−0.5 percent

at 24 months with 6 and 12 lags). Fourth, two measures of the trend in the two-year yield work less well. With a market-based measure of the nominal neutral rate—the 9–10-year forward rate from the Federal Reserve's fitted yield curve (Gürkaynak et al., 2007), which replaces both the expected-inflation and r^* adjustments—unemployment rises with any of the FCIs (by up to 0.3 percentage point, significant mostly at the 68 percent level) but the price level barely falls; distant forward rates contain a large and time-varying term premium, which fell over the sample. With the two-year yield measured relative to its trailing 8-year average, the nominal and real FCIs give conventional unemployment responses but no significant price response, and the r^*-gap FCI gives weak responses; the 8-year average and the Laubach–Williams trend are close until 2015 and diverge after 2016, when the 8-year average fell to about 1 percent while the Laubach–Williams trend stayed near 3–4 percent. Overall, removing a slow-moving neutral-rate component from the two-year yield matters, model-based estimates of r^* (Laubach–Williams or Kiley) and a smooth deterministic trend do this well, and backward-looking averages and distant forward rates do it poorly.

Table 4: Alternative trends in rates: unemployment gap (top) and core PCE price level (bottom)

		12 months			24 months			36 mon	
Rate	FCI	3	6	12	3	6	12	3	6
2-year r^*-gap, Laubach–Williams (baseline)	none	0.08	0.03	−0.08	0.16†	0.04	−0.15†	0.20†	0.05
2-year r^*-gap, Laubach–Williams (baseline)	nominal	0.26**	0.23**	0.21**	0.38**	0.32**	0.28†	0.35**	0.31*
2-year r^*-gap, Laubach–Williams (baseline)	real	0.31**	0.28**	0.26**	0.43**	0.39**	0.34**	0.37**	0.36*
2-year r^*-gap, Laubach–Williams (baseline)	r^*-gap	0.30**	0.32**	0.34**	0.38**	0.39**	0.41**	0.34**	0.38*
2-year r^*-gap, Kiley	none	0.00	−0.11	−0.22**	0.06	−0.10	−0.29**	0.09	−0.0
2-year r^*-gap, Kiley	nominal	0.18**	0.13†	0.12†	0.27**	0.22†	0.19†	0.25**	0.22†
2-year r^*-gap, Kiley	real	0.24**	0.19**	0.16†	0.33**	0.29**	0.24†	0.27**	0.27*
2-year r^*-gap, Kiley	r^*-gap	0.29**	0.27**	0.27**	0.37**	0.37**	0.35**	0.32**	0.35*
2-year − 9–10y forward rate	none	−0.23**	−0.22**	−0.26**	−0.33**	−0.24†	−0.35**	−0.34**	−0.22
2-year − 9–10y forward rate	nominal	0.09†	0.10	0.09	0.15†	0.21†	0.14	0.16†	0.22†
2-year − 9–10y forward rate	real	0.17**	0.18†	0.16†	0.22**	0.29**	0.24†	0.21**	0.27*
2-year − 9–10y forward rate	r^*-gap	0.07	0.17†	0.25**	0.04	0.20†	0.29†	0.03	0.15†
2-year nominal + quadratic trend	none	0.07	−0.13	−0.27**	0.36†	0.10	−0.19	0.53**	0.29
2-year nominal + quadratic trend	nominal	0.44**	0.36**	0.28**	0.76**	0.75**	0.63**	0.76**	0.83*
2-year nominal + quadratic trend	real	0.41**	0.32**	0.25†	0.69**	0.67**	0.57**	0.68**	0.74*
2-year nominal + quadratic trend	r^*-gap	0.39**	0.31**	0.29**	0.58**	0.57**	0.55**	0.52**	0.57*
2-year − 8-year average	none	−0.22**	−0.30**	−0.36**	−0.23†	−0.28†	−0.46**	−0.22	−0.2
2-year − 8-year average	nominal	0.23**	0.18†	0.15†	0.38**	0.36**	0.27†	0.34**	0.36*
2-year − 8-year average	real	0.25**	0.19**	0.15†	0.37**	0.36**	0.28†	0.31**	0.34*
2-year − 8-year average	r^*-gap	0.06	0.08	0.14†	0.10†	0.15†	0.18†	0.08	0.15†

		12 months			24 months			36 mo	
Rate	FCI	3	6	12	3	6	12	3	6
2-year r^*-gap, Laubach–Williams (baseline)	none	−0.15†	−0.13†	−0.19**	−0.22†	−0.26†	−0.31†	−0.28†	−0.3
2-year r^*-gap, Laubach–Williams (baseline)	nominal	−0.18**	−0.22**	−0.25**	−0.32**	−0.45**	−0.47**	−0.38†	−0.6
2-year r^*-gap, Laubach–Williams (baseline)	real	−0.20**	−0.24**	−0.29**	−0.34†	−0.49**	−0.55**	−0.40†	−0.66
2-year r^*-gap, Laubach–Williams (baseline)	r^*-gap	−0.18†	−0.25**	−0.33**	−0.28†	−0.48**	−0.61**	−0.37†	−0.68
2-year r^*-gap, Kiley	none	−0.13†	−0.07	−0.12†	−0.18	−0.11	−0.11	−0.22	−0.1
2-year r^*-gap, Kiley	nominal	−0.16†	−0.15†	−0.15†	−0.26†	−0.30†	−0.22†	−0.31†	−0.4
2-year r^*-gap, Kiley	real	−0.19†	−0.18†	−0.20**	−0.30†	−0.35†	−0.31†	−0.35†	−0.4
2-year r^*-gap, Kiley	r^*-gap	−0.18†	−0.21**	−0.26**	−0.28†	−0.40†	−0.42†	−0.35†	−0.5
2-year − 9–10y forward rate	none	0.13†	0.01	−0.04	0.32**	0.14	0.07	0.42**	0.23
2-year − 9–10y forward rate	nominal	0.03	−0.09	−0.12	0.06	−0.10	−0.12	0.06	−0.1
2-year − 9–10y forward rate	real	0.01	−0.12	−0.18†	0.05	−0.14	−0.24	0.07	−0.1
2-year − 9–10y forward rate	r^*-gap	0.04	−0.18†	−0.28**	0.20	−0.17	−0.42†	0.30	−0.1
2-year nominal + quadratic trend	none	−0.03	−0.01	−0.02	−0.16	−0.16	−0.07	−0.26†	−0.3
2-year nominal + quadratic trend	nominal	−0.09	−0.11	−0.11	−0.24	−0.36†	−0.31†	−0.33	−0.5
2-year nominal + quadratic trend	real	−0.09	−0.11	−0.13	−0.23	−0.34†	−0.33	−0.31	−0.5
2-year nominal + quadratic trend	r^*-gap	−0.20**	−0.22**	−0.21**	−0.32**	−0.50**	−0.52**	−0.32†	−0.6
2-year − 8-year average	none	0.19†	0.10	0.02	0.37†	0.22	0.10	0.50†	0.28
2-year − 8-year average	nominal	0.01	−0.07	−0.11	−0.02	−0.19	−0.23	−0.02	−0.2
2-year − 8-year average	real	0.02	−0.07	−0.14†	0.02	−0.15	−0.28†	0.05	−0.1
2-year − 8-year average	r^*-gap	0.07	−0.05	−0.15†	0.19	−0.07	−0.27†	0.29	−0.0

See Table 1. FCI column: the FCI version used. The FCI is always one of the companion note's three indexes; rows differ only in the adjustment of the two-year yield. The 8-year average and Laubach–Williams trends are close until 2015 (averages 5.3 and 5.5 percent over 1994–2007; 2.3 and 2.5 over 2008–2015) and diverge after 2016 (1.4 and 3.6).

The pandemic. The baseline drops March–December 2020 as dependent observations, but 2020 values still enter the 2021 equations as lags. Table 5 also drops 2021 and, separately, ends the sample in December 2019. Both changes weaken the results, and the price response depends on the post-2020 data. With the r^*-gap rate and the FCI, dropping 2021 lowers the unemployment response to 0.16–0.26 percentage point at 12 months (significant at the 90 percent level with 3 and 12 lags) and about 0.15–0.25 at 24–36 months (significant mostly at the 68 percent level); the price level still falls significantly with 12 lags (−0.42 percent at 24 months) but not with 3 or 6. Ending the sample in 2019, unemployment rises 0.20–0.30 at 12 months (significant at the 90 percent level with 3 and 12 lags) and 0.17–0.30 later (mostly significant at the 68 percent level), while the price level barely moves. Without the FCI, unemployment is insignificant or negative in both samples. The unemployment result is therefore robust to the treatment of the pandemic in sign and, at short horizons, in significance, but smaller; the significant price response comes largely from the period since 2021, when inflation rose and fell with a large swing in the stance of policy relative to r^*.

Table 5: Sample: unemployment gap (top) and core PCE price level (bottom), r^*-gap rate

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Baseline sample	r^*-gap (LW)	0.30**	0.32**	0.34**	0.38**	0.39**	0.41**	0.34**	0.38**	0.37**
Also drop 2021	r^*-gap (LW)	0.16**	0.17†	0.26**	0.18†	0.16	0.25†	0.15†	0.13	0.16†
End in 2019	r^*-gap (LW)	0.20**	0.23†	0.30**	0.21†	0.21†	0.30**	0.17	0.17	0.23†
Baseline sample	none	0.08	0.03	−0.08	0.16†	0.04	−0.15†	0.20†	0.05	−0.16
Also drop 2021	none	−0.05	−0.11	−0.16†	−0.06	−0.18	−0.30**	−0.06	−0.20	−0.37**
End in 2019	none	−0.09	−0.09	−0.14	−0.13	−0.14	−0.19	−0.15	−0.15	−0.16

		12 months			24 months			36 months		
Rate	FCI	3	6	12	3	6	12	3	6	12
Baseline sample	r^*-gap (LW)	−0.18†	−0.25**	−0.33**	−0.28†	−0.48**	−0.61**	−0.37†	−0.68**	−0.84**
Also drop 2021	r^*-gap (LW)	0.05	−0.09	−0.23**	0.16	−0.11	−0.42**	0.18	−0.16	−0.62**
End in 2019	r^*-gap (LW)	−0.04	−0.07†	−0.05	0.00	−0.04	−0.12	0.03	−0.04	−0.21†
Baseline sample	none	−0.15†	−0.13†	−0.19**	−0.22†	−0.26†	−0.31†	−0.28†	−0.39	−0.44†
Also drop 2021	none	0.01	0.01	−0.06	0.08	0.01	−0.11	0.14	−0.01	−0.20
End in 2019	none	0.02	0.04	0.10**	0.08†	0.08	0.10†	0.13†	0.10	0.07

See Table 1. "End in 2019" also estimates the daily block through 2019.

Other checks. In exploratory versions of the model, adding a Baa–Treasury spread to the daily block lowered the unemployment response by a third to a half; because no intraday credit series exists, the spread's window response could not be measured cleanly, and we do not pursue it here, but credit is the most important omitted channel (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019). Replacing the model's daily dynamics with the directly estimated post-announcement path of the financial variables for the first one to two months gives larger but much less precise responses.

5 Conclusion

High-frequency surprises identify monetary policy shocks well; the difficulty has been in tracing their effects on the economy. Holding the surprise fixed and changing only its propagation, we find that two standard features of the transmission mechanism account for much of the post-1994

puzzle. Policy works through financial conditions as a whole, and it is the level of rates relative to a slow-moving neutral rate that matters for spending and inflation. Propagating window-identified surprises through a broad financial conditions index, with rates measured relative to the Laubach–Williams r^*, gives conventional, precise and lag-robust responses of unemployment and prices of a size comparable to the best recent estimates.

The evidence is strongest for the role of financial conditions, which matters in every variant. Removing the slow-moving component of rates matters for precision; for unemployment, simple trends do about as well as r^*; Kiley's r^* and a smooth trend also deliver price responses, while a backward-looking average and a market-based proxy, the distant forward rate, do not. The significant price response depends on a model-based estimate of r^* (Laubach–Williams or Kiley), which embeds a model of the inflation-relevant neutral rate but is itself estimated from macroeconomic data, or a smooth trend, and on the post-2021 period; before the pandemic the unemployment response is conventional but the price response is small. These findings complement the information-effect and predictability literatures; the mechanisms can operate together, and combining orthogonalized or decomposed surprises with our propagation is a natural next step. The main limitations are the identifying assumption—that policy reaches the economy through the financial variables in the model, and that the economy responds to them the same way whatever moved them—the omission of credit spreads, and intervals that do not include uncertainty about r^* or the daily dynamics. The broader lesson is that the decline in r^* over the past three decades is not only a fact about the level of rates but also a measurement problem for estimating the effects of policy.

References

Acosta, M., A. Ajello, M. Bauer, F. Loria and S. Miranda-Agrippino (2025), "Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database," Federal Reserve Bank of San Francisco Working Paper 2025-30.

Ajello, A., M. Cavallo, G. Favara, W. Peterman, J. Schindler and N. Sinha (2023), "A New Index to Measure U.S. Financial Conditions," FEDS Notes, Board of Governors of the Federal Reserve System, June 30.

Bauer, M. D. and G. D. Rudebusch (2020), "Interest Rates under Falling Stars," American Economic Review 110(5), 1316–1354.

Bauer, M. D. and E. T. Swanson (2023a), "A Reassessment of Monetary Policy Surprises and High-Frequency Identification," NBER Macroeconomics Annual 2022, Vol. 37, 87–155.

Bauer, M. D. and E. T. Swanson (2023b), "An Alternative Explanation for the 'Fed Information Effect'," American Economic Review 113(3), 664–700.

Bernanke, B. S. and K. N. Kuttner (2005), "What Explains the Stock Market's Reaction to Federal Reserve Policy?" Journal of Finance 60(3), 1221–1257.

Boivin, J., M. T. Kiley and F. S. Mishkin (2010), "How Has the Monetary Transmission Mechanism Evolved Over Time?" in B. M. Friedman and M. Woodford (eds.), Handbook of Monetary Economics, Vol. 3A, 369–422.

Caldara, D. and E. Herbst (2019), "Monetary Policy, Real Activity, and Credit Spreads: Evidence from Bayesian Proxy SVARs," American Economic Journal: Macroeconomics 11(1), 157–192.

Chudik, A. and G. Georgiadis (2022), "Estimation of Impulse Response Functions When Shocks Are Observed at a Higher Frequency Than Outcome Variables," Journal of Business and Economic Statistics 40(3), 965–979.

Chudik, A. and L. Kilian (2026), "Mean Group and Pooled Mixed-Frequency Estimators of Responses of Low-Frequency Variables to High-Frequency Shocks," Federal Reserve Bank of Dallas Working Paper 2603.

Companion note (2026), "A Simple Market-Based Financial Conditions Index: Weights from Policy News, Dynamics from a Transparent VAR, Measured Relative to r^*," draft.

Faust, J., E. T. Swanson and J. H. Wright (2004), "Identifying VARs Based on High Frequency Futures Data," Journal of Monetary Economics 51(6), 1107–1131.

Francis, N., E. Ghysels and M. T. Owyang (2011), "The Low-Frequency Impact of Daily Monetary Policy Shocks," Federal Reserve Bank of St. Louis Working Paper 2011-009.

Gazzani, A. and A. Vicondoa (2020), "Bridge Proxy-SVAR: Estimating the Macroeconomic Effects of Shocks Identified at High-Frequency," Banca d'Italia Working Paper 1274.

Gertler, M. and P. Karadi (2015), "Monetary Policy Surprises, Credit Costs, and Economic Activity," American Economic Journal: Macroeconomics 7(1), 44–76.

Ghysels, E. (2016), "Macroeconomics and the Reality of Mixed Frequency Data," Journal of Econometrics 193(2), 294–314.

Gilchrist, S. and E. Zakrajšek (2012), "Credit Spreads and Business Cycle Fluctuations," American Economic Review 102(4), 1692–1720.

Gürkaynak, R. S., B. Sack and E. T. Swanson (2005), "Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements," International Journal of Central Banking 1(1), 55–93.

Gürkaynak, R. S., B. Sack and J. H. Wright (2007), "The U.S. Treasury Yield Curve: 1961 to the Present," Journal of Monetary Economics 54(8), 2291–2304.

Hamilton, J. D., E. S. Harris, J. Hatzius and K. D. West (2016), "The Equilibrium Real Funds Rate: Past, Present, and Future," IMF Economic Review 64(4), 660–707.

Hanson, S. G. and J. C. Stein (2015), "Monetary Policy and Long-Term Real Rates," Journal of Financial Economics 115(3), 429–448.

Hatzius, J., P. Hooper, F. Mishkin, K. Schoenholtz and M. Watson (2010), "Financial Conditions Indexes: A Fresh Look after the Financial Crisis," NBER Working Paper 16150.

Haubrich, J., G. Pennacchi and P. Ritchken (2012), "Inflation Expectations, Real Rates, and Risk Premia: Evidence from Inflation Swaps," Review of Financial Studies 25(5), 1588–1629.

Holston, K., T. Laubach and J. C. Williams (2017), "Measuring the Natural Rate of Interest: International Trends and Determinants," Journal of International Economics 108, S59–S75.

Jarociński, M. and P. Karadi (2020), "Deconstructing Monetary Policy Surprises—The Role of Information Shocks," American Economic Journal: Macroeconomics 12(2), 1–43.

Kiley, M. T. (2020), "What Can the Data Tell Us about the Equilibrium Real Interest Rate?" International Journal of Central Banking 16(3), 181–209.

Kilian, L. (2024), "How to Construct Monthly VAR Proxies Based on Daily Surprises in Futures Markets," Journal of Economic Dynamics and Control 168, 104966.

Kuttner, K. N. (2001), "Monetary Policy Surprises and Interest Rates: Evidence from the Fed Funds Futures Market," Journal of Monetary Economics 47(3), 523–544.

Laubach, T. and J. C. Williams (2003), "Measuring the Natural Rate of Interest," Review of Economics and Statistics 85(4), 1063–1070.

Miranda-Agrippino, S. and G. Ricco (2021), "The Transmission of Monetary Policy Shocks," American Economic Journal: Macroeconomics 13(3), 74–107.

Nakamura, E. and J. Steinsson (2018), "High-Frequency Identification of Monetary Non-Neutrality: The Information Effect," Quarterly Journal of Economics 133(3), 1283–1330.

Ramey, V. A. (2016), "Macroeconomic Shocks and Their Propagation," in J. B. Taylor and H. Uhlig (eds.), Handbook of Macroeconomics, Vol. 2A, 71–162.

Romer, C. D. and D. H. Romer (2000), "Federal Reserve Information and the Behavior of Interest Rates," American Economic Review 90(3), 429–457.

Swanson, E. T. (2021), "Measuring the Effects of Federal Reserve Forward Guidance and Asset Purchases on Financial Markets," Journal of Monetary Economics 118, 32–53.